



OPEN
Compute Project

Hardware Fault Management Project
System Debug Specification

0.1

Modular Base Specification
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1 Acknowledgements

We wish to acknowledge Chris Fenner (Google) and the Trusted Computing Group for the use of their Pandoc markdown tooling.

2 Compliance with OCP Tenets

2.1 Openness

This specification is open.

2.2 Efficiency

This specification is efficient.

2.3 Impact

This specification is impactful.

2.4 Scale

This specification is scalable.

2.5 Sustainability

This specification is sustainable.

3 Introduction

4 Definitions

4.1 Life cycle

5 CodeCurrency

More often than not the root cause of a failure or problem is simply that the System has not been configured correctly, it is run either outdated FW, an incompatible combination of SW/FW components or components that have not been configured appropriately. The concept of CodeCurrency has been designed to address this issue by providing a 1 stop shop style standardized machine readable method to retrieve all relevant information about the configuration of a given system. The information will be exposed by a system as a file in the Redfish profile of the System. Components and subsystem will expose their part of the information to the Main BMC as PLDM type 7 files. The content of the File is organized using the JSON schema outlined in this file.

5.0.1 Node CodeCurrency Record Schema

This section defines the **JSON schema** for representing the firmware, configuration, and physical/logical details of nodes, assemblies, and components in a datacenter environment.

5.0.1.1 Schema

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "Node Codecurency Record",
  "type": "object",
  "properties": {
    "NodeId": { "type": "string", "description": "Unique identifier for
    ↳ the node" },
    "NodeType": { "type": "string", "description": "Type of node (e.g.,
    ↳ AI Server, Ethernet Switch, Management Tray)" },
    "Hostname": { "type": "string", "description": "Node hostname" },
    "NetworkConfiguration": {
      "type": "object",
      "description": "Network configuration details for the node",
      "properties": {
        "IpAddresses": { "type": "array", "items": { "type": "string",
        ↳ "format": "ipv4" } },
        "Ports": { "type": "array", "items": { "type": "integer",
        ↳ "minimum": 1, "maximum": 65535 } },
        "MacAddresses": { "type": "array", "items": { "type": "string",
        ↳ "pattern": "^[0-9A-Fa-f]{2:}{5}[0-9A-Fa-f]{2}$" } }
      },
      "required": ["IpAddresses"]
    },
    "PhysicalLocation": {
      "type": "object",
      "description": "Physical location of the node in the datacenter",
      "properties": {
        "Rack": { "type": "string" },
        "Row": { "type": "string" },
        "Datacenter": { "type": "string" },
        "Placement": {
          "type": "object",
          "properties": {
            "Type": { "type": "string", "enum": ["RackSlot", "Bay"] },

```

```

        "Slot": { "type": "integer", "description": "Rack unit
        ↪ position if RackSlot" },
        "HeightInU": { "type": "integer", "description": "Height in U
        ↪ if RackSlot" },
        "BayId": { "type": "string", "description": "Bay identifier
        ↪ if Bay" },
        "BayPosition": { "type": "string", "description": "Position
        ↪ within the bay" }
    },
    "required": ["Type"]
}
},
"required": ["Rack", "Row", "Datacenter", "Placement"]
},
"Assemblies": {
    "type": "array",
    "description": "List of assemblies within the node (supports nested
    ↪ assemblies)",
    "items": { "$ref": "#/$defs/Assembly" }
}
},
"required": ["NodeId", "NodeType", "Assemblies"],
"$defs": {
    "Assembly": {
        "type": "object",
        "properties": {
            "AssemblyName": { "type": "string" },
            "AssemblyId": { "type": "string" },
            "ManufacturerSerialNumber": { "type": "string", "description":
            ↪ "Manufacturer's serial number for the assembly" },
            "PhysicalLocation": {
                "type": "object",
                "description": "Physical slot or bay where this assembly is
                ↪ installed",
                "properties": {
                    "Slot": { "type": "string", "description": "Physical slot
                    ↪ identifier (e.g., PCIe slot, OCP slot)" },
                    "Position": { "type": "string", "description": "Position
                    ↪ within the node or assembly" }
                }
            },
        },
    },
    "LogicalAssociation": {
        "type": "object",
        "description": "Logical connectivity details for this
        ↪ assembly",
        "properties": {
            "Links": {
                "type": "array",
                "description": "List of logical links consumed by this
                ↪ assembly",
                "items": {
                    "type": "object",
                    "properties": {

```

```

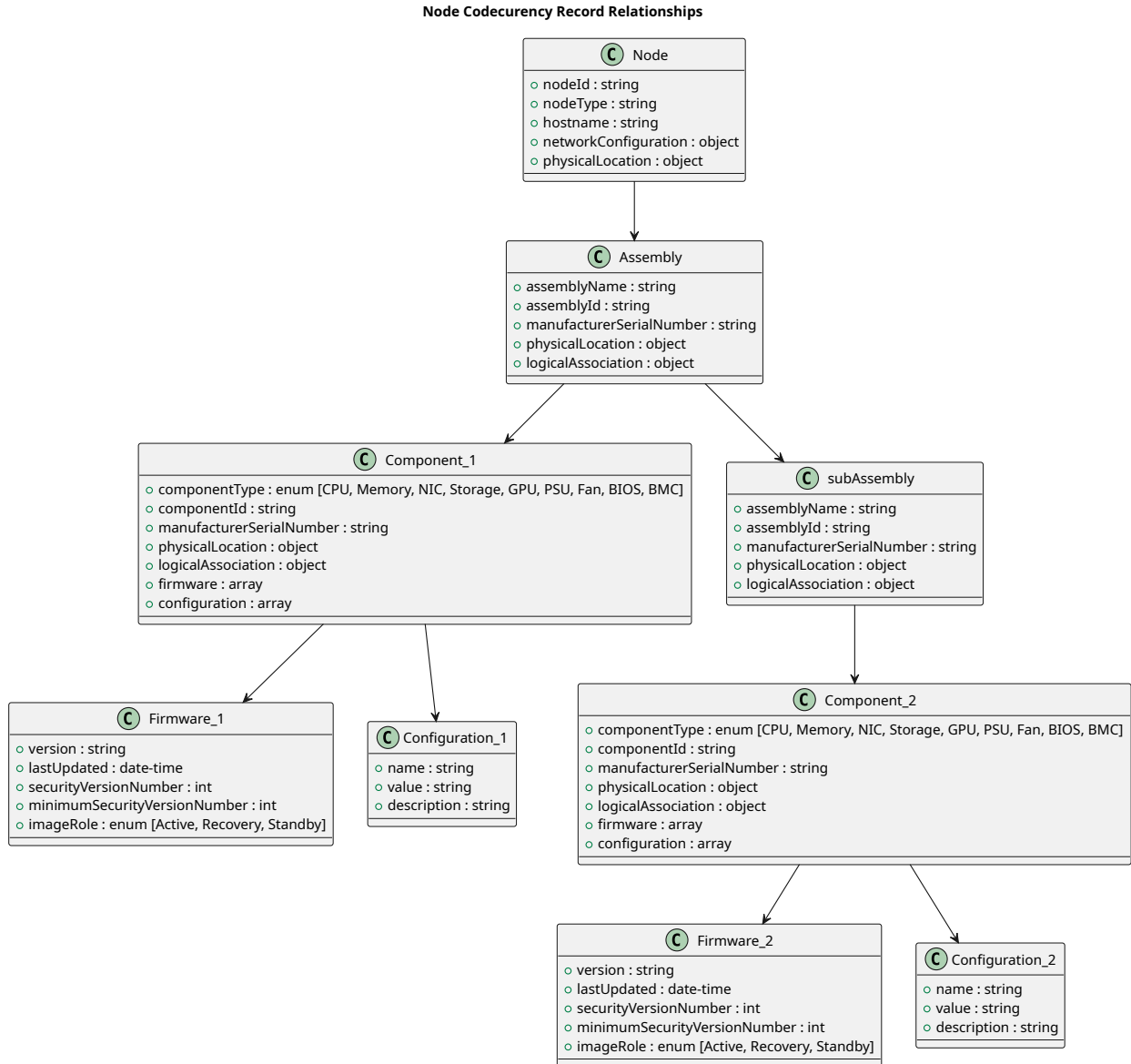
        "ConnectedTo": { "type": "string" },
        "Protocol": { "type": "string" },
        "LaneCount": { "type": "integer" }
    },
    "required": ["ConnectedTo", "Protocol"]
},
}
}
},
"Components": { "type": "array", "items": { "$ref":
    ↪ "#/$defs/Component" } },
"SubAssemblies": { "type": "array", "items": { "$ref":
    ↪ "#/$defs/Assembly" } }
},
"required": ["AssemblyName", "AssemblyId"]
},
"Component": {
    "type": "object",
    "properties": {
        "ComponentType": { "type": "string", "enum": ["CPU", "Memory",
            ↪ "NIC", "Storage", "GPU", "PSU", "Fan", "BIOS", "BMC", "AMC"]
            ↪ },
        "ComponentId": { "type": "string" },
        "ManufacturerSerialNumber": { "type": "string", "description":
            ↪ "Manufacturer's serial number for the component" },
        "PhysicalLocation": {
            "type": "object",
            "description": "Physical slot or bay where this component is
                ↪ installed",
            "properties": {
                "Slot": { "type": "string", "description": "Physical slot
                    ↪ identifier (e.g., PCIe slot, DIMM slot)" },
                "Position": { "type": "string", "description": "Position
                    ↪ within the node or assembly" }
            }
        },
    },
    "LogicalAssociation": {
        "type": "object",
        "description": "Logical connectivity details for this
            ↪ component",
        "properties": {
            "Links": {
                "type": "array",
                "description": "List of logical links consumed by this
                    ↪ component",
                "items": {
                    "type": "object",
                    "properties": {
                        "ConnectedTo": { "type": "string" },
                        "Protocol": { "type": "string" },
                        "LaneCount": { "type": "integer" }
                    },
                    "required": ["ConnectedTo", "Protocol"]
                }
            }
        }
    }
}

```

```

    }
  }
},
"Firmware": {
  "type": "array",
  "description": "List of firmware entries for the component",
  "items": {
    "type": "object",
    "properties": {
      "Version": { "type": "string", "description": "Firmware
↪ version" },
      "LastUpdated": { "type": "string", "format": "date-time",
↪ "description": "Timestamp of last firmware update" },
      "SecurityVersionNumber": { "type": "integer",
↪ "description": "Current Security Version Number (SVN)"
↪ },
      "MinimumSecurityVersionNumber": { "type": "integer",
↪ "description": "Minimum enforced Security Version
↪ Number (SVN)" },
      "ImageRole": {
        "type": "string",
        "enum": ["Active", "Recovery", "Standby"],
        "description": "Indicates if this firmware is actively
↪ running or a redundant recovery image"
      }
    },
    "required": ["Version"]
  }
},
"Configuration": {
  "type": "array",
  "description": "Custom configuration attributes for the
↪ component",
  "items": {
    "type": "object",
    "properties": {
      "Name": { "type": "string", "description": "Configuration
↪ attribute name" },
      "Value": { "type": "string", "description": "Configuration
↪ attribute value" },
      "Description": { "type": "string", "description": "Optional
↪ description of the attribute" }
    },
    "required": ["Name", "Value"]
  }
},
"required": ["ComponentType", "ComponentId", "Firmware"]
}
}

```

**Figure 1:** Codecurency relationships

5.0.1.2 Field Explanations

5.0.1.2.1 Node-Level

- **nodeId**: Unique identifier for the node.
- **nodeType**: Type of node (e.g., AI Server, Ethernet Switch).
- **hostname**: Hostname for network identification.
- **networkConfiguration**:
 - **ipAddresses**: List of IPv4 addresses.
 - **ports**: Open or configured ports.
 - **macAddresses**: MAC addresses for network interfaces.

- **physicalLocation:**
 - **rack, row, datacenter:** Physical placement identifiers.
 - **placement:**
 - * **type:** RackSlot or Bay.
 - * **slot:** Rack unit position.
 - * **heightInU:** Height in rack units.
 - * **bayId, bayPosition:** Bay-specific identifiers.

5.0.1.2.2 Assemblies

- **assemblyName:** Name of the assembly (e.g., Baseboard).
- **assemblyId:** Unique identifier.
- **manufacturerSerialNumber:** Manufacturer's serial number.
- **physicalLocation:** Slot and position within node.
- **logicalAssociation:**
 - **links:** Array of logical connections (connectedTo, protocol, laneCount).
- **components:** Array of components.
- **subAssemblies:** Nested assemblies.

5.0.1.2.3 Components

- **componentType:** Type (CPU, Memory, NIC, etc.).
- **componentId:** Unique identifier.
- **manufacturerSerialNumber:** Manufacturer's serial number.
- **physicalLocation:** Slot and position.
- **logicalAssociation:** Logical links consumed by the component.
- **firmware:**
 - **version:** Firmware version (required).
 - **lastUpdated:** Timestamp of last update.
 - **securityVersionNumber:** Current SVN.
 - **minimumSecurityVersionNumber:** Minimum enforced SVN.
 - **imageRole:** Active, Recovery, or Standby.
- **configuration:**
 - Array of custom attributes: **name**, **value**, optional **description**.

5.0.1.3 Example

Below is an example of an Hypothetical Node with :

- **Dual-socket CPUs**
- **Memory modules**
- **BMC**
- **Two NVMe drives**
- **Two NICs on a hot-swap backplane**
- **UBB subassembly with BMC**
- **Two OAM subassemblies** (each with GPU and AMC)

```
{
  "NodeId": "AI-Node-001",
  "NodeType": "AI Server",
  "Hostname": "ai-node-001.dc.local",
  "NetworkConfiguration": {
    "IpAddresses": ["192.168.10.101"],
    "Ports": [22, 443],
    "MacAddresses": ["00:1A:2B:3C:4D:5E"]
  },
  "PhysicalLocation": {
    "Rack": "R42",
    "Row": "Row-A",
    "Datacenter": "DC-West",
    "Placement": {
      "Type": "RackSlot",
      "Slot": 12,
      "HeightInU": 2
    }
  },
  "Assemblies": [
    {
      "AssemblyName": "Baseboard",
      "AssemblyId": "BB-001",
      "ManufacturerSerialNumber": "BB-SN-12345",
      "Components": [
        {
          "ComponentType": "CPU",
          "ComponentId": "CPU-001",
          "ManufacturerSerialNumber": "CPU-SN-11111",
          "Firmware": [
            {
              "Version": "1.2.3",
              "LastUpdated": "2025-10-01T12:00:00Z",
              "SecurityVersionNumber": 5,
              "MinimumSecurityVersionNumber": 3,
              "ImageRole": "Active"
            }
          ]
        },
        {
          "ComponentType": "CPU",
          "ComponentId": "CPU-002",
          "ManufacturerSerialNumber": "CPU-SN-22222",
          "Firmware": [
            {
              "Version": "1.2.3",
              "LastUpdated": "2025-10-01T12:00:00Z",
              "SecurityVersionNumber": 5,
              "MinimumSecurityVersionNumber": 3,
              "ImageRole": "Active"
            }
          ]
        }
      ]
    }
  ]
}
```

```
    },
    {
      "ComponentType": "Memory",
      "ComponentId": "MEM-001",
      "ManufacturerSerialNumber": "MEM-SN-33333",
      "Firmware": [
        { "Version": "N/A" }
      ]
    },
    {
      "ComponentType": "Memory",
      "ComponentId": "MEM-002",
      "ManufacturerSerialNumber": "MEM-SN-44444",
      "Firmware": [
        { "Version": "N/A" }
      ]
    },
    {
      "ComponentType": "BMC",
      "ComponentId": "BMC-001",
      "ManufacturerSerialNumber": "BMC-SN-55555",
      "Firmware": [
        {
          "Version": "3.4.5",
          "LastUpdated": "2025-09-15T08:00:00Z",
          "SecurityVersionNumber": 2,
          "MinimumSecurityVersionNumber": 1,
          "ImageRole": "Active"
        }
      ]
    }
  ],
  "SubAssemblies": [
    {
      "AssemblyName": "HotSwap Backplane",
      "AssemblyId": "HSBP-001",
      "ManufacturerSerialNumber": "HSBP-SN-66666",
      "Components": [
        {
          "ComponentType": "Storage",
          "ComponentId": "NVME-001",
          "ManufacturerSerialNumber": "NVME-SN-77777",
          "Firmware": [
            { "Version": "2.1.0" }
          ]
        },
        {
          "ComponentType": "Storage",
          "ComponentId": "NVME-002",
          "ManufacturerSerialNumber": "NVME-SN-88888",
          "Firmware": [
            { "Version": "2.1.0" }
          ]
        }
      ]
    }
  ]
}
```

```
    },
    {
      "ComponentType": "NIC",
      "ComponentId": "NIC-001",
      "ManufacturerSerialNumber": "NIC-SN-99999",
      "Firmware": [
        { "Version": "1.0.0" }
      ]
    },
    {
      "ComponentType": "NIC",
      "ComponentId": "NIC-002",
      "ManufacturerSerialNumber": "NIC-SN-10101",
      "Firmware": [
        { "Version": "1.0.0" }
      ]
    }
  ],
},
{
  "AssemblyName": "UBB",
  "AssemblyId": "UBB-001",
  "ManufacturerSerialNumber": "UBB-SN-20202",
  "SubAssemblies": [
    {
      "AssemblyName": "OAM-1",
      "AssemblyId": "OAM-001",
      "ManufacturerSerialNumber": "OAM-SN-30303",
      "Components": [
        {
          "ComponentType": "GPU",
          "ComponentId": "GPU-001",
          "ManufacturerSerialNumber": "GPU-SN-40404",
          "Firmware": [
            { "Version": "510.85" }
          ]
        },
        {
          "ComponentType": "AMC",
          "ComponentId": "AMC-001",
          "ManufacturerSerialNumber": "AMC-SN-50505",
          "Firmware": [
            { "Version": "1.0.0" }
          ]
        }
      ]
    }
  ],
},
{
  "AssemblyName": "OAM-2",
  "AssemblyId": "OAM-002",
  "ManufacturerSerialNumber": "OAM-SN-60606",
  "Components": [
```

```

"ComponentType": "GPU",
"ComponentId": "GPU-002",
"ManufacturerSerialNumber": "GPU-SN-70707",
"Firmware": [
  { "Version": "510.85" }
],
},
{
"ComponentType": "AMC",
"ComponentId": "AMC-002",
"ManufacturerSerialNumber": "AMC-SN-80808",
"Firmware": [
  { "Version": "1.0.0" }
]
}
]
}
]
}
]
}

```

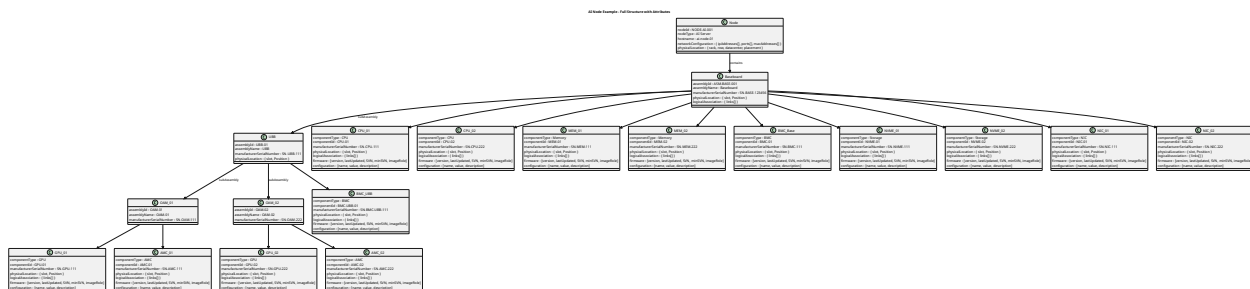


Figure 2: Example Codecurrency

5.0.1.4 CodeCurrency Redfish schema

6 Status Dump

7 Future work

Appendix A: First appendix

This is the first appendix.

Appendix B: Second appendix

This is the second appendix.

References